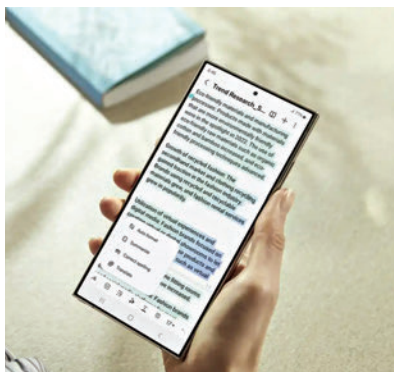


those are just the most visible applications. Its footprint extends across a diverse array of form factors, including wearables, automotive systems, and IoT devices. This widespread adoption has been made possible by leaps in AI algorithms and hardware optimisation, allowing even the most compact devices to boast AI capabilities. This democratisation of AI technology signifies a future where intelligence is embedded in every digital interaction, from the way we drive to how we live and communicate.

## THE PERKS

The benefits of on-device AI extend far beyond mere convenience. Enhanced privacy, reduced latency, and offline functionality stand out as hallmark benefits. By processing data locally, these AI models ensure that sensitive information never leaves the device, offering a fortress of privacy in an era of increasing data breaches. Moreover, the elimination of cloud dependency means instant responses to user commands, transforming user experience into something seamless and intuitive. As it stands right now, instant responses are a far cry since we're looking at the first iterations of AI models that directly respond to the user. The term 'near-instant' is more appropriate. Nevertheless, these benefits are not just theoretical; they manifest tangibly across all devices, making technology not just faster and safer, but also more aligned with human needs.

True democratisation cannot happen with proprietary AI models. Every manufacturer wants to keep their AI refinements close to their hearts. The rise of open-source AI models is going to be the true game-changer in the democratisation of AI technology. By making cutting-edge models accessible to all, they lower the barrier to entry for developers and small companies, sparking a wave of innovation that was previously confined to tech giants. This open ecosystem fosters a collaborative environment where advancements are shared, not hoarded, propelling the entire industry forward and ensuring that the benefits of AI are widespread and equitable.



## THE CHALLENGES

Despite its promise, the road to ubiquitous on-device AI is fraught with challenges. Hardware limitations, energy consumption, and ethical dilemmas such as bias and privacy concerns are significant hurdles. Not a day goes by without a news article showcasing how certain AI models used copyrighted information to train their models. In fact, several major corporations are holding back on the mass adoption of popular AI models for this very concern. You wouldn't want to issue a recall or patch out an AI feature because you had later discovered that it was trained on data obtained illegally. Especially, not when you've charged a premium for the 'AI features'. The industry giants such as Google, Qualcomm, Samsung, and OpenAI are leading several initiatives in this domain, including collaboration with industry partners and investment in ethical AI research, highlight a comprehensive approach to overcoming these obstacles, ensuring that on-device AI can scale responsibly and sustainably. You'll still come across the occasional article or two stating that even these biggies have inadvertently used data that shouldn't have been but such instances will go down. Especially since artists have started fighting back with poisoning tools that embed tiny pieces of data that mess up the AI models which train on them.

## WHERE ELSE?

From healthcare monitoring via wearable devices to smart home automation and real-time language translation on smartphones, the applications of on-device AI

are as diverse as they are transformative. Not everyone's bodies are the same. The polymorphism of certain genes across races among human beings results in different responses to certain medicines. So healthcare is one of the domains which has a huge potential for personalisation. Your wearables are going to become much smarter in the coming years to enable personalised care.

While self-driving cars have been a reality for a while now, not all cars come with the same smarts. That's changing thanks to the Advanced driver-assistance systems (ADAS) that companies such as Qualcomm have been pioneering. Real-time decision making, automatic braking, maintaining lane discipline and detecting pedestrians are just some of the smart features creeping into premium cars. It won't be long before they're present even in the cheapest cars that you can buy. After all, the hardware enabling these services isn't that cost prohibitive anymore.

Smart home devices such as thermostats, lights, security cameras have also started incorporating AI models, albeit not as smart as the ones in our phones. Portable diagnostic devices are also seeing AI models being built into them. Patients with chronic conditions can be alerted way in advance and ailments can be detected way before they become chronic. Our life expectancies can only go up with such smart health monitoring devices. Whether it's providing critical health insights on the go or breaking down language barriers in real time, on-device AI is reshaping our world in profound ways.

## THE NEW NORMAL

The impact of on-device AI extends far beyond mere technological innovation; it heralds a new chapter in our relationship with technology - a new normal where devices act as extensions of our selves, understanding and anticipating our needs while safeguarding our privacy. As we stand on the brink of this new era, it's clear that on-device AI is not just transforming our devices; it's reshaping our world, making technology more personal, intelligent, and, ultimately, human. 